Finite Groups and Their Representations — Exercise Sheet 5

Representations and invariant complements

Variant: Questions only

Exercise 1 (Averaging a non-equivariant projection). Let S_3 permute the coordinates of $V = \mathbb{C}^3$, and put $u = (1,1,1)$ and $L = \mathbb{C}u$. The map $\pi(x) = x_1 u$ is a projection onto L. Compute each conjugate $\rho(g)\pi\rho(g)^{-1}$ and their average π_0. Find its kernel and explain why that kernel is an invariant complement to L.

Exercise 2 (The dihedral family). For $n \geq 3$ put $\zeta = \exp(2\pi \mathrm{i}/n)$. For $k \in \{0, \ldots, n-1\}$ put

$$R_k = \begin{pmatrix} \zeta^k & 0 \\ 0 & \zeta^{-k} \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(i) For $0 \leq a < n$ and $b \in \{0,1\}$, show that $\rho_k(r^a s^b) = R_k^a S^b$ is a representation of D_n. Prove it is irreducible exactly when $2k \not\equiv 0 \pmod{n}$. Describe its splitting otherwise.

(ii) For irreducible members prove $\rho_k \cong \rho_l$ exactly when $l \equiv k$ or $-k \pmod{n}$.

(iii) Prove that every irreducible complex representation of D_n is one-dimensional or one of these planes. List the one-dimensional ones and count all irreducibles, without using character orthogonality.

(iv) The square acts on $\mathbb{R}^2$ by the rotation r and reflection s with matrices

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Over $\mathbb{C}$, put $u = (1, -\mathrm{i})$ and use the basis (u, su) to identify this geometric representation with ρ_1 for $n = 4$.

Exercise 3 (Representations of a product of cyclic groups). Let $A = C_{m_1} \times \cdots \times C_{m_t}$, with each $m_i \geq 2$ and specified generators a_i. Include $t = 0$, when A is trivial. List all irreducible complex representations up to isomorphism. Write $\mathrm{Irr}(A)$ for a set of representatives of these isomorphism classes, and prove $|\mathrm{Irr}(A)| = |A|$. For $A = C_4 \times C_6$, give the formulas explicitly and decide whether any irreducible representation is faithful.

Exercise 4 (Intertwiners and their possible ranks). For S_3, let $E = \{x \in \mathbb{C}^3 : x_1 + x_2 + x_3 = 0\}$ be the irreducible plane of Lecture 5. Put

$$V = \mathbf{1}^{\oplus 2} \oplus E^{\oplus 3} \oplus \mathrm{sgn}, \qquad W = \mathbf{1} \oplus E^{\oplus 2} \oplus \mathrm{sgn}^{\oplus 4}.$$

Find $\dim \mathrm{Hom}_{S_3}(V, W)$, $\dim \mathrm{End}_{S_3}(V)$ and the largest possible rank of an intertwiner $V \to W$. Can an intertwiner be injective or surjective?

Exercise 5 (Why the hypotheses of Maschke matter).

(i) Let the infinite cyclic group $\langle g\rangle \cong \mathbb{Z}$ act on $\mathbb{C}^2$ by $\rho(g) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Show that $\mathbb{C}e_1$ has no invariant complement, although the field has characteristic zero. Compute $\rho(g^n)$ for every $n \in \mathbb{Z}$.

(ii) Let the finite group C_2 act on the abelian group $\mathbb{Z}^2$ by interchanging coordinates. Show that the invariant subgroup $\mathbb{Z}(1,1)$ has no invariant direct-sum complement in $\mathbb{Z}^2$. Compare with the corresponding vector space over $\mathbb{Q}$. Here a complement means an additive subgroup whose direct sum with $\mathbb{Z}(1,1)$ is $\mathbb{Z}^2$.

Exercise 6 (An invariant inner product in coordinates). Let $\zeta = \exp(2\pi \mathrm{i}/3)$ and let C_3 act on $V = \mathbb{C}^2$ with generator matrix

$$R = \begin{pmatrix} \zeta & 1-\zeta \\ 0 & 1 \end{pmatrix}.$$

Verify $R^3 = \mathbb{1}$. Starting with the standard inner product, compute the matrix $H = \frac{1}{3}\sum_{j=0}^{2} (R^j)^\dagger R^j$ of its invariant average. Find the averaging projection onto the line V^{C_3} of invariant vectors, and verify that its kernel is orthogonal to that line for the new inner product.

Exercise 7 ⋆. Prove the classification of finite abelian groups into products of non-trivial cyclic prime-power groups, uniquely up to ordering, with this guided construction. The trivial group is the empty product.

(i) For a non-trivial abelian p-group A, choose a of largest order $m = p^f$. Start with $\chi : \langle a\rangle \to \mu_m$, sending a to a primitive m-th root, where $\mu_m = \{z \in \mathbb{C} : z^m = 1\}$. For a subgroup $H \leq A$ and $x \in A$, show that any homomorphism $\chi : H \to \mu_m$ extends to $H\langle x\rangle$: let d be the least positive integer with $x^d \in H$ and choose $\eta \in \mu_m$ with $\eta^d = \chi(x^d)$.

(ii) Extend to all of A and prove $A = \langle a\rangle \times \ker\chi$. Iterate, and combine the Sylow factors for a general finite abelian group.

(iii) If $A \cong \prod_i C_{p^{f_i}}$ with positive integers f_i, prove that the numbers

$$r_j = \log_p \left|\{x \in A : x^{p^j} = e\}\right|, \qquad j \geq 0,$$

determine the multiset of f_i. For $C_8 \times C_4 \times C_2$, compute r_0, r_1, r_2, r_3.